

Personal Statements

Brian

My fascination with biomedical engineering, especially its entrepreneurial aspects, has been a driving force throughout my academic journey. From a young age, I had ideas for startups—most of which were more imaginative than practical. But when my interests shifted toward medical innovations, the potential to make a meaningful impact became real. I've been interested in studying stress in pain points since I took running seriously. I delved into the world of shoe design to optimize comfort and ergonomics. When I saw an idea for babies with large omphaloceles to be more comfortable, I was fascinated.

I'm a designer at my core, and that has allowed me to crosslink different interests in my life to solve problems. I believe in pushing boundaries and creating things that didn't exist before, and I'm always on the lookout for new ways to express myself. Curiosity drives everything I do. I'm constantly exploring different mediums and cultural influences, blending architecture, medicine, and engineering to tell a story that's deeper and multi-dimensional. CHOC's Tummy Time project follows my philosophy.

This project represents a step toward my goal of exploring the entrepreneurial side of biomedical engineering. While I have a solid foundation in biomedical engineering through academic coursework and research, I haven't yet had the chance to apply these skills independently. I look forward to gaining experience in quality assurance and learning what it takes to produce impactful medical devices at scale. I am particularly drawn to the potential redesign aspect of the project, which I hope will allow me to contribute innovative ideas to improve the device's efficiency and quality.

In terms of background, I took a design course, "BME 60C: Engineering Analysis/Design: Computer-Aided Design," where I led a team project that highlighted my ability to delegate tasks, maintain team accountability, and support decision-making while fostering

trust and independence. These skills were further developed in a team competition I participated in from July to October 2024. This experience solidified my leadership capabilities and gave me a taste of what it's like to see a project through from start to finish.

Isaac

I've learned a lot in the first several years of my biomedical engineering degree, but not until this year did I appreciate the directive to address patient needs with biomedical engineering. Guest speakers in my senior design class highlighted examples of engineering having a real clinical impact by patching the scenarios where existing medical solutions are inadequate. The Tummy Time project with CHOC offers an opportunity to address a meaningful unmet need for infants born with large omphaloceles. I know that I can make a difference by using my engineering skills to enable the safe prone positioning of infants with omphaloceles.

Although it has a small market, this medical need is incredibly durable. Spending time on their stomachs, even for only a few minutes a day, is hugely beneficial for infants and their development. Some infants, who already suffer from a congenital health challenge, cannot safely reap the benefits that tummy time confers to so many avenues of infant development. I want to help them, and I know that I'm up to the task.

This project aligns with my larger goals as an engineer. It combines creative problem-solving with the technical challenges of developing a safe and functional product for the most sensitive patients. I hope to leverage the understanding of stress mechanics and materials design that I've gleaned from my biomedical engineering coursework to come up with a mattress device that adequately supports an infant, without interfering with its omphaloceles whatsoever.

The Tummy Time project represents more than a design challenge to me; it represents an opportunity to make a lasting difference for infants and their families. By creating a solution for a unique clinical need, I aim to improve the quality of life and development of these infants and offer their families peace of mind. I want to apply my technical knowledge by addressing the blind spots of modern medical care. As I work toward a career where I can continue addressing unmet medical needs, this project serves as a meaningful step in preparing me for that path.